Sound Ring

Fortnite's "visualize sound effects" — in real life, for under $100

The idea

In Fortnite, a ring around your character shows where sounds are coming from and what they are (footsteps, gunshots, chests). We build the same thing as a small wearable: a ring of LEDs on a hat or headband that lights up toward a sound and changes color based on the type of sound (siren = red, dog = orange, doorbell = blue, speech = green, etc.). Useful for people who are deaf or hard of hearing, and just a fun build.

How it works

- **Four microphones** on one small board hear the sound at slightly different times. Comparing those timing differences tells us the direction (same trick your two ears use).
- **A machine-learning model (YAMNet)** listens to the same audio and names the sound. It's free from Google and already knows about 500 sounds.
- **An LED ring** lights the LED nearest the direction, colored by the sound type.
- Everything runs on a Raspberry Pi strapped to the hat, powered by a USB battery pack. Updates about twice per second.

Parts list (about $60–75)

Part	Price	Why
Raspberry Pi Zero 2 W	~$15	The computer
ReSpeaker 4-Mic Array HAT (Seeed)	~$25–30	4 mics + 12 RGB LEDs on one board, plugs onto the Pi
microSD card (16 GB+)	~$6	Storage / operating system
USB power bank	~$10–15	A few hours of battery
Headband or cap, velcro/zip ties	~$5	Mount the board flat on top
Foam mic windscreen (optional)	~$5	Helps a lot outdoors

No soldering. The mic board has the LEDs built in, so it's two boards, a card, and a battery.

Build steps

- Flash Raspberry Pi OS onto the SD card and boot the Pi.
- Plug the ReSpeaker HAT onto the Pi and install its driver (one script).
- Install Python packages and download the YAMNet model (free).
- Run `sound_ring.py`. Clap from the front once to calibrate which LED is "forward."
- Mount it on a hat or headband, mics facing up. Done.

What to expect

- Direction is good (±15–30°) for sharp sounds like knocks, barks, claps, sirens. Fuzzier for hums and distant traffic.

- It shows direction, not distance. Brightness can stand in for loudness.
- About half a second from sound to light.
- Indoors, echoes sometimes make it point at a wall. Works best in open rooms and outdoors.
- It works while walking; all the mics are on your head, so "to my right" always means your right.

Upgrades later

- Add a $3 motion sensor (IMU) so the light stays locked on a sound as you turn your head.
- Train a custom model (Edge Impulse, free) on the 5–10 sounds we actually care about — faster and more accurate.
- Move a short LED strip inside glasses frames for a glance-level cue at the edge of your vision.

Code and full setup notes: `sound_ring.py` and `README.md`.